

Architecture & Design Overview

A senior/junior credit-tranche vault layered on the Wildcat v -wmtUSDC market wrapper. This describes the system as built: its contracts, how it attaches to a Wildcat market, the design decisions, and the properties it guarantees.

Ethereum mainnet · full rationale in the Design & Risk Specification

CAPITAL STACK	FIRST-LOSS	SENIOR LEVERAGE	DEFAULT	VALUATION
Senior + Junior	Junior ≥ 20%	≤ 4×	grace + 90d	realised-only

1 Contracts

CONTRACT	ROLE
WaterfallMath	Pure tranche math: senior target accrual, value/loss split, subordination, and the ToU default trigger.
TrancheController	Immutable orchestrator: deposits, the async redemption queue, default detection & wind-down, subordination gating, per-user sanctions + escrow. Reentrancy-guarded; safe transfers.
senior / junior	sr-wmtUSDC / jr-wmtUSDC: Solady ERC20 + EIP-2612 permit with an ERC-4626 value-view surface. Senior open to KYC'd lenders; junior whitelisted.
TrancheFactory	Registered at the WildcatArchController; deploys one tranche set per registered market, gated on isRegisteredMarket.
IExternal	Interfaces to the Wildcat 4626 wrapper, market (state + withdrawal queue), sanctions sentinel, and arch controller.

2 How it attaches



The TrancheFactory is registered at the protocol level (WildcatArchController) and deploys one senior/junior set per registered market. Tranching is a lender-side vault on top of an unchanged market; the borrower continues to see a single unitranche facility.

- **Senior** holds a priority claim that accrues at a target derived from the facility's base APR (the APR times a governance-set, timelocked senior share, capped at the APR), funded from realised yield and topped from junior down to a floor. It is a priority claim, not a solvency guarantee.
- **Junior** is the leveraged first-loss residual: it earns the excess spread and absorbs losses to zero before senior is touched.
- **Subordination** is a fixed floor (junior $\geq 20\%$ of TVL, senior $\leq 4\times$), enforced by gating senior deposits and junior withdrawals.
- **Valuation** is realised-only and oracle-free: NAV is held at a high-watermark while the market is delinquent, so unrealised penalty accrual is never booked.
- **Redemption** is asynchronous, mirroring the market's batched withdrawal queue (request $\rightarrow$ execute $\rightarrow$ claim); partial proceeds are allocated senior-first, and while the market is delinquent or winding down junior may only draw settled cash beyond the *entire* senior obligation, so first-loss capital cannot exit ahead of senior during a slow-motion default. Exits during delinquency are sized at the live price, so frozen-mark appreciation stays in the pool. Recovery crediting is balance-derived with a permissionless sync (), so cash arriving outside the poke path is never stranded.
- **Default** mirrors Wildcat ToU §6.2 on-chain (`timeDelinquent  $\geq$  delinquencyGracePeriod + 90` days), per-market configurable, with a Loan-Agreement override. On trigger the vault freezes deposits, halts senior accrual, and winds down senior-first.
- **Governance** is immutable logic + a bounded, timelocked parameter set; pause halts deposits only and never blocks senior exits.
- **Compliance** re-enforces the Wildcat sentinel per user (sanctioned redemptions route to escrow); junior is restricted to qualified providers.

④ Properties & verification

INVARIANTS

`seniorValue + juniorValue == realisedValue` at all times · junior floors at zero before senior is impaired (first-loss) · senior is paid before junior on every settlement (priority).

- Built on Solady's audited ERC20, ReentrancyGuard, and SafeTransferLib; share conversions round in favour of the pool.
- Covered by a Foundry suite of 55 tests: unit/behaviour, property fuzzing, 128k-call stateful invariants (conservation, junior-first-loss, no over-distribution), and mainnet-fork tests that exercise deposit, valuation, and a redemption round-trip against the live `v-wmtUSDC` facility and its withdrawal queue. Review regressions are pinned in `AuditPoC.t.sol`, each stating the finding it covers.

Two independent three-pass agentic review cycles have been run against the code, with every actionable finding fixed and regression-pinned; no human audit engagement has been performed yet — one precedes any real capital. Independent review focuses on the credit-loss / redemption edge cases, the gas/DoS profile of the redemption-request array at scale, and ERC-4626 conformance of the `USDC` sub-protocol.